

Participant 33 — T0 method (label-free)

Method, code and theory behind `submissions/Results_33_T0_0.csv` (PPP_WTG17) and `submissions/Results_33_T0_final.csv` (SSS_WTG06). Both files come from the same model with no fitted parameter:

$$\hat{\theta}(d) = L - (\bar{r}_{\ell(d)} - \alpha_{\text{seg}(\ell(d))})$$

Symbol	Meaning	Source
$\hat{\theta}(d)$	predicted static yaw misalignment on day d	
$\ell(d)$	the day's state; submitted as <code>cluster</code>	change points of r (§3)
$\bar{r}_{\ell}$	level of the neighbour heading residual in state ℓ	§2
α_{seg}	anchor: day-weighted mean of $\bar{r}_{\ell}$ over an encoder-free segment	§4
L	level prior: the turbine's vane set-point	§4

In words: the *changes* of the misalignment are read from how the nacelle heading moves against its neighbours, with a slope of exactly -1 that follows from the sensor algebra; the *level* is taken from the vane set-point.

1. Sensor model

Unobserved: true wind direction W , true nacelle heading N , yaw error $\gamma = W - N$ (wrapped to $\pm 180^\circ$). The SCADA stream reports

$$N_r = N + \delta_n, \quad v = g\gamma + \delta_v, \quad W_r = N_r + v.$$

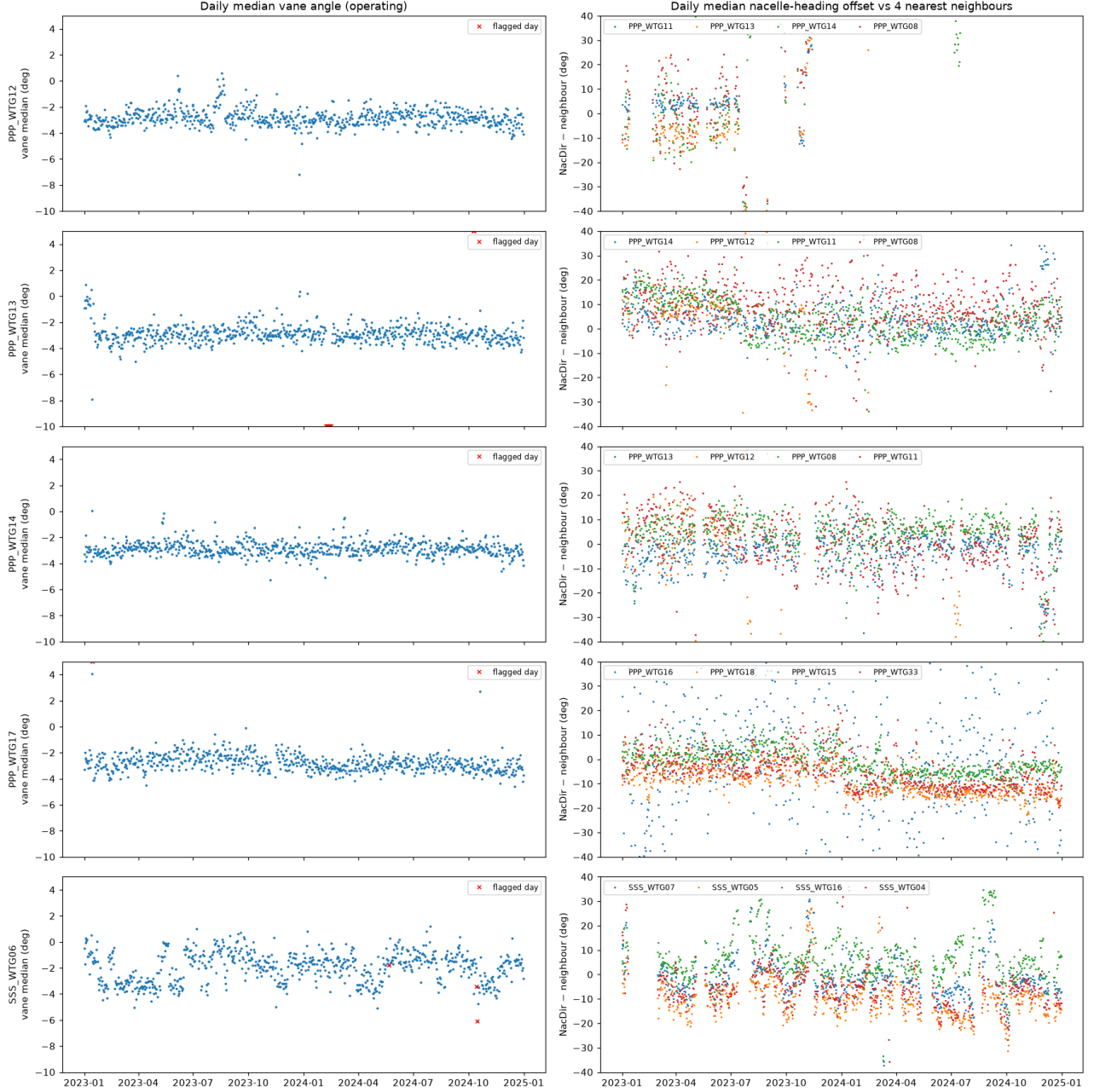
- δ_n : the nacelle encoder has no north reference. It is an arbitrary constant that jumps whenever the encoder is re-referenced.
- v : the vane reads the yaw error with a mounting offset δ_v and a gain $g \leq 1$ (we use $g = 1$; it cannot be measured from this data).
- The third equation is how the data set is encoded, not an assumption: the vane is recovered as `vane = wrap(WindDir - NacDir)`, so `WindDir` carries the encoder offset and only the vane is encoder-free.

The yaw controller rotates the nacelle until the filtered vane sits at its set-point s , so in steady state $E[v] = s$ and

$$\theta \equiv E[\gamma] = \frac{s - \delta_v}{g}.$$

θ is the static misalignment: the persistent yaw error the controller cannot see, because it sits inside its own measurement.

Consequence: the vane cannot see θ . Averaged over a day the vane reads s whatever θ is. The left column below shows it: the daily median vane is flat at about -3° on every PPP turbine. Information about θ has to come from an external direction reference, and the neighbours provide one (right column: heading differences against the four nearest turbines, before any cleaning).



2. The neighbour heading residual

Let $c = W + \kappa$ be a consensus of the wind direction from the neighbours, with κ a constant made of the neighbours' own offsets. Then

$$N_r - c = N + \delta_n - W - \kappa = -\gamma + \delta_n - \kappa,$$

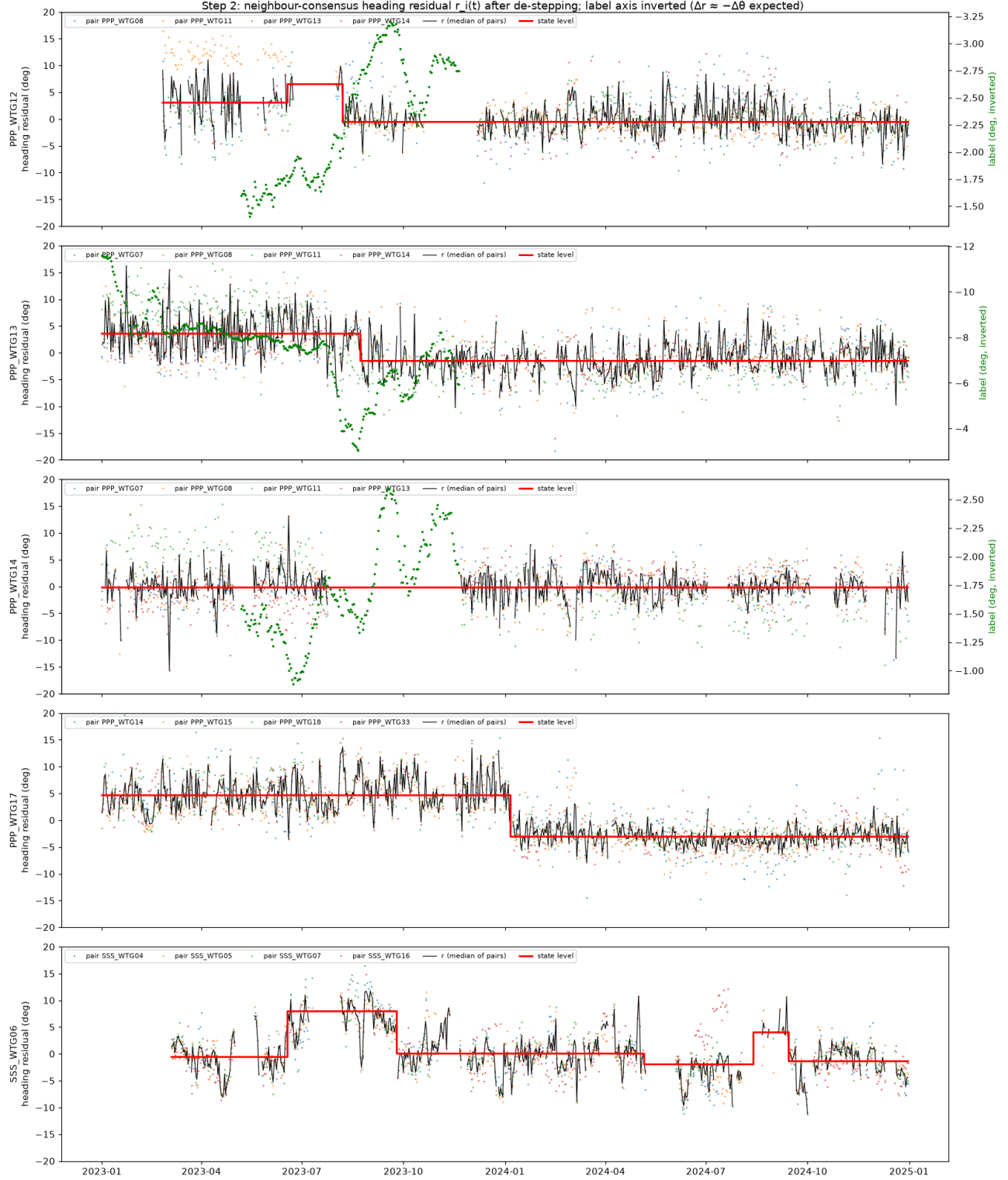
and averaged over a day

$$r(d) = -\theta(d) + \delta_n(d) - \kappa \quad \implies \quad \Delta r = -\Delta\theta \quad \text{whenever } \Delta\delta_n = 0.$$

So the residual gives **changes** of the misalignment in physical degrees, with slope -1 , independent of the vane and of any power model. Its **level** is not identifiable, because δ_n and κ are unknown constants.

How r is built (10-minute aggregates, circular statistics throughout):

1. **Pair series.** For turbine i and each of its 4 nearest same-site neighbours j : $D_{ij} = \text{wrap}(N_r^i - N_r^j)$, in bins where both turbines operate.
2. **Encoder frames.** A shift of i 's encoder frame moves D_{ij} against *all* neighbours by the same amount on the same day. Jumps of at least 25° with the same sign in at least 60 % of the pairs within ± 3 days are credited to the encoder and cut the record into intervals. Most of them are not re-references but **excursions**: the heading reads about 100° off for two weeks and returns. Subtracting an estimated jump out and another one back leaves their difference (1° to 30°) in r , so no jump estimate is subtracted for them:
 - intervals shorter than 45 days are excursions and are dropped, with ± 3 days around every step;
 - between two consecutive long intervals the shift is measured locally (first 40 days after minus last 40 days before, median over neighbours). Below 15° the frame came back and nothing is corrected; above it the shift is a re-reference, is subtracted, and marks an encoder boundary.
3. **Transient removal.** Days more than 10° from the pair's rolling median over 91 days with data are dropped. A rolling median keeps a level shift of any size but not a pulse shorter than half the window, so this removes the short excursions, of either turbine, that are too small to be credited in step 2. The price: a real state shorter than about 45 days that departs by more than 10° would be removed too.
4. **Sector correction.** Terrain and wakes make the heading difference depend on wind direction, so the median of D_{ij} per 30° sector of the *neighbour's* wind direction is subtracted. This absorbs the neighbour's own offsets into κ .
5. **Robust daily value.** Median over the bins of the day per pair, then median over pairs: $r_i(d)$. Day-to-day noise is about 2° .



Coloured dots are the individual pairs, the black line is r , the red line the state levels of §3. On the three training turbines the label is drawn in green on an inverted axis, so r and the label should move together.

3. States

r is smoothed with a centred 7-day rolling median and segmented with PELT (squared-error cost, penalty 400, minimum state length 14 days). The penalty means a jump J lasting n days on both sides is accepted when $nJ^2/2 \gtrsim 400$, e.g. 2.4° over 140 days or 5° over 30 days. Each state gets the median of r as its level $\bar{r}_\ell$.

A boundary with a jump above 15° , or within 10 days of a re-reference verified in step 2, is typed as an **encoder** boundary. r is not comparable across such a boundary.

The states are computed from the SCADA residual over all 731 days, with no knowledge of the template or of which days are scored. They are the **cluster** column.

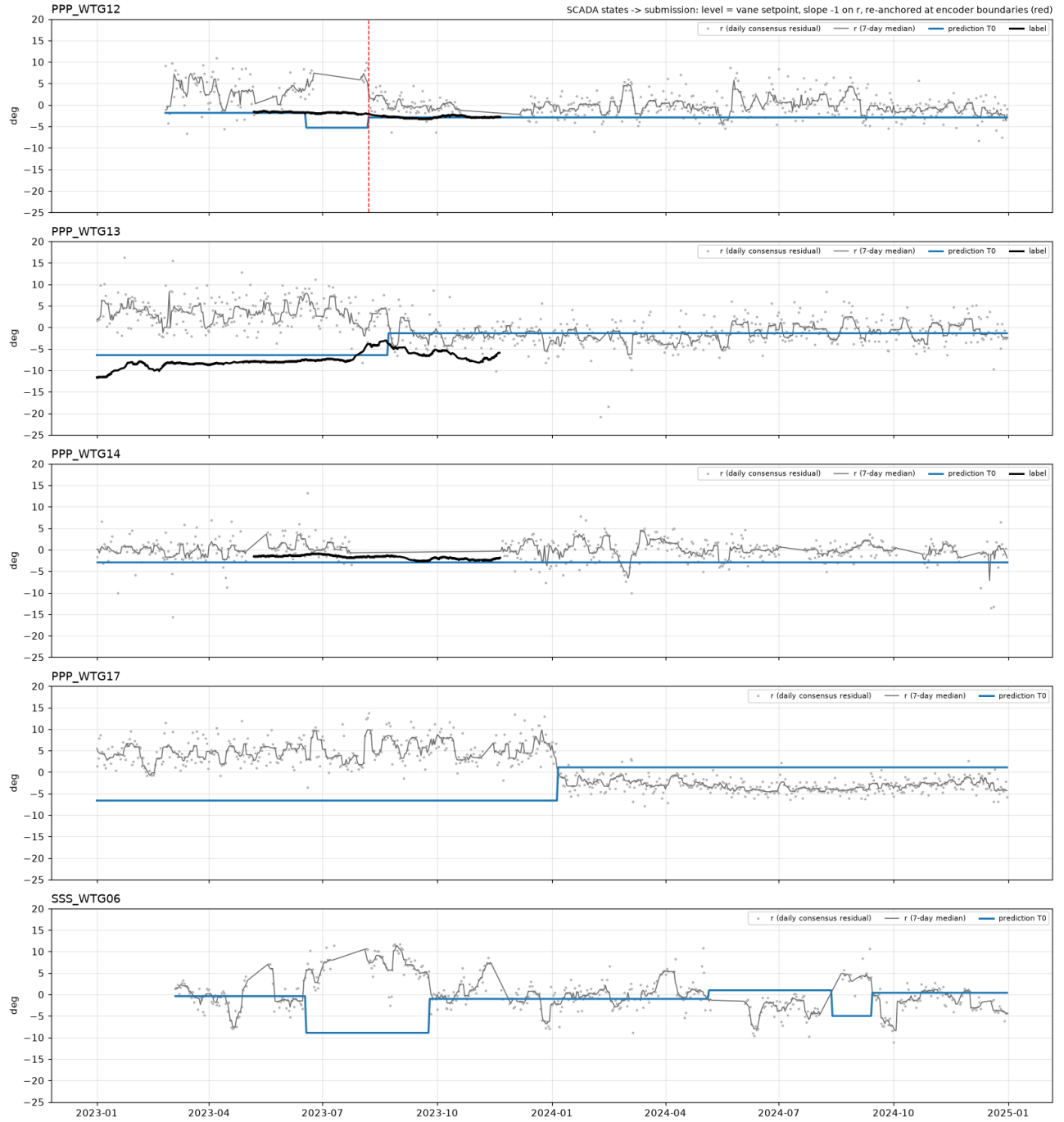
4. Level: vane set-point and anchor

No label-free estimator of the absolute level survived testing. Locating the peak of the power-versus-yaw curve fails because over the controller's $\pm 8^\circ$ deadband the power changes by only about 2 % ($\cos^2$ model), the same order as sector, turbulence and season effects.

The level is therefore a prior. Assuming an unbiased vane ($\delta_v = 0$, $g = 1$), $\theta = s$, and s is measured as

$$L = \text{median}_{d \text{ healthy}} (\text{daily median vane in operation}).$$

The prior describes the turbine's *typical* state, so the state levels are centred before the slope is applied. Within each encoder-free segment (a run of states with no encoder boundary) the anchor α is the day-weighted mean of the state levels. Re-anchoring per segment stops an undetected re-reference from leaking into the predictions.



Blue is the prediction, grey the residual it is derived from, black the label where it exists (training turbines only). Red dashed lines are encoder boundaries.

5. What was submitted

File	Turbine	States	Predicted levels
Results_33_T0_0.csv	PPP_WTG17	2 (boundary 2024-01-06, $\Delta r = -7.74^\circ$)	-6.6° then $+1.1^\circ$
Results_33_T0_final.csv	SSS_WTG06	6, in 1 encoder-free segment	$-8.9^\circ \dots +1.0^\circ$, mean -1.6°

The six levels on SSS_WTG06 are -0.4° (to 2023-06-17), -8.9° (to 2023-09-24), -1.0° (to 2024-05-05), $+1.0^\circ$ (to 2024-08-12), -5.0° (to 2024-09-13) and $+0.4^\circ$.

The validate file merged on the leaderboard was written by the first version of the method, before steps 2 and 3 of §2 were changed (-6.8° then $+1.3^\circ$). The file in `submissions/` is the output of the present code and differs from it by at most 0.16° .

6. Use of labels

No parameter of the model is fitted to labels: the slope is -1 from §2 and the level is the vane set-point. The training labels were used for two things only.

- **Sign check.** The script refuses to write a file unless the label moves opposite to r on the training turbines (measured slope -1.07), which fixes the sign convention of θ .
- **Evaluation.** Scored on the three training turbines with the challenge rule, the model reaches RMSE 2.18° , MAE 1.64° , bias $+0.20^\circ$, ARI 0.36. Segmentation thresholds were set while these turbines, with their labels, were in view.

7. Known limits

- **The level is a prior, not a measurement.** It is wrong by $-\delta_v$ when the vane itself is offset. On the training turbines the bias is about 1° on two (-1.0° , -1.2°) and $+2.3^\circ$ on the third.
- **A small re-reference looks like a misalignment change.** Below 15° the two cannot be told apart in r , so boundaries are more reliable than magnitudes.
- **Steps below about 1° are under the noise floor of r .**
- **SSS_WTG06 is the uncertain case.** Its encoder frame is in an excursion on 117 of 731 days and the site has only four neighbours, two of which have excursions of their own. The first 63 days have no residual and take the value of the first state. The three states of 2024 differ by 1° to 6° and are the least certain part of the file.
- **A short, large state is removed as a transient.** Anything shorter than about 45 days that departs by more than 10° and returns is treated as an encoder excursion (§2 step 3).

8. Outlook: fleet use and further work

The method needs nacelle heading and power from the standard SCADA stream of a turbine and a few neighbours, no labels and no extra sensor, and runs in about two minutes for 16 turbines over two years. On both held-out turbines it found the large changes ($8-9^\circ$) to within a day or two, with the step size in physical degrees. That makes it a candidate for **fleet screening**: “the alignment of this turbine changed by about X° around this date”, which triggers an inspection. It is not a replacement for a measurement campaign.

Weaknesses for a fleet roll-out

- **No absolute level.** The residual sees changes only (§2). The level is the vane set-point prior, which is 2.3° off on PPP_WTG13 and cannot be checked without a label. A turbine misaligned by a constant amount since commissioning is invisible. Closing this with one calibration per turbine is exactly the cost a label-free method is meant to avoid.
- **Noise floor of several degrees.** Day-to-day noise of r is $1-2^\circ$, but r also wanders by up to $\pm 5^\circ$ over weeks. Changes below $3-5^\circ$ are not reliable. The training labels only contain changes of $1-3^\circ$, so the range where the method works rests on two events (PPP_WTG17, SSS_WTG06) whose labels we do not have.
- **Dependence on healthy neighbours.** The consensus is a median over four pairs. On SSS_WTG06 two of the four neighbours had excursions of their own in August–September 2024, and the file has a 32-day state at -5.0° there that is probably not real. Small sites, edge turbines, curtailed or stopped neighbours are weak spots; an isolated turbine cannot be assessed.
- **Encoder events cost data and can hide a change.** SSS_WTG06 loses 117 of 731 days to excursions and has a residual on 519 days. A change during an excursion is dated to its end, and one of 15° or more across an excursion is removed as a re-reference. A service visit that touches both vane and encoder, the likeliest real cause of a misalignment change, is exactly the confounded case.

- **Short states are removed.** Anything shorter than about 45 days that departs by more than 10° and returns is deleted, including a real fault of that shape (a loose vane fixed after a month).
- **Hand-set thresholds.** 25° , 15° , 10° , 45 days, 91 days and the PELT penalty were set on five turbines of one type at two sites. The first version failed on the test site for this reason: the validate site never exercised the de-stepping. Other controllers, encoders and terrain will break some of them.
- **Sector correction assumes a stationary site.** Sector medians are taken over the whole record; seasonal stability, vegetation, new neighbouring turbines or a changed curtailment strategy shift heading differences without any misalignment.
- **Retrospective.** Centred windows of 7 and 91 days and a minimum state length of 14 days mean a change is confirmed one to two months later.
- **Changes of the vane gain are invisible** to the residual.

Further work

1. **A level estimate.** Combine the residual with a within-turbine, encoder-free level estimator such as the power response during yaw manoeuvres. It is noisy per day, but only one number per state is needed, and the states of §3 say which days may be pooled. Alternatively, find out how few spot calibrations per farm suffice when the residual carries the level between turbine-states.
2. **Farm-wide solution.** Solve all pair series of a site jointly, with one piecewise-constant offset per turbine and robust weights, instead of a median over four fixed neighbours. This makes the attribution of a shift to a turbine explicit and removes the manual reference blacklist.
3. **Segmentation on valid days with a noise model.** PELT currently runs on a series interpolated across masked days with a fixed penalty. Valid days only and a penalty scaled to the turbine's own noise would remove the weak 2024 states on SSS_WTG06 for a stated reason, not by tuning.
4. **Uncertainty per state.** An interval for every $\Delta\theta$ from the spread across pairs and days, and a flag when a boundary coincides with an excursion of the turbine or a neighbour.
5. **Use the encoder events.** Excursion and re-reference dates are a by-product, maintenance-relevant in themselves, and mark the days where a real change is most likely and least observable.
6. **Validation on more labelled turbines.** Labelled changes of $3\text{--}10^\circ$, a second turbine type and a site with fewer than four neighbours are the missing cases.
7. **Online operation.** One-sided windows, detection delay as a function of step size, and a false-alarm rate measured on turbines known to be stable.